SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C04 - ASSESSMENT TOOL 3 – Viva Voice Questions

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 – Viva Voice Questions**	**Assessment:**	**Assessment Tool 3 – Viva Based**
Weightage:	**20% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyse and apply statistical inference, frequentist approaches, point estimation, variability measures, confidence intervals, and hypothesis testing techniques to draw conclusions about population parameters and interpret statistical results effectively. (BL4)		
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Section / Batch:	AI&DS	**Date:**	21-08-2026

Code of Conduct

I Yedururi Anjani Priya (192324176) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Anjani Priya

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	

		reasoning and insights				
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

A. Introduction to Statistical Inference

1. What is statistical inference?

Statistical inference is the process of using sample data to make conclusions or predictions about a larger population.

2. Why is statistical inference important in data science?

It helps data scientists make decisions about a population without collecting data from every individual, saving time, cost, and resources.

3. What is the difference between descriptive statistics and inferential statistics?

- **Descriptive statistics** summarize and describe the available data.
- **Inferential statistics** use sample data to make conclusions about a population.

4. What is a population in statistics?

A population is the complete set of individuals, objects, or observations being studied.

5. What is a sample?

A sample is a smaller subset selected from a population for analysis.

6. Why do data scientists usually work with samples instead of complete populations?

Because collecting data from an entire population can be expensive, time-consuming, difficult, or sometimes impossible.

7. What is a population parameter?

A population parameter is a numerical value that describes a characteristic of an entire population, such as population mean ((\mu)) or population proportion ((p)).

8. What is a sample statistic?

A sample statistic is a numerical value calculated from sample data, such as sample mean ((\bar{x})) or sample proportion ((\hat{p})).

9. Give a real-world example where statistical inference is used in industry.

An e-commerce company may survey 500 customers and use their responses to estimate the satisfaction level of all customers.

10. What are the two major activities involved in statistical inference?

The two major activities are:

1. **Estimation**
2. **Hypothesis testing**

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

The frequentist approach uses sample data and probability based on repeated experiments or long-run frequencies to estimate population parameters and test hypotheses.

12. What is the basic idea behind frequentist statistics?

Probability is interpreted as the long-run frequency of an event occurring over many repeated trials.

13. How does the frequentist approach differ from the Bayesian approach?

- **Frequentist:** Parameters are fixed and data are random.
- **Bayesian:** Parameters can be treated as uncertain and updated using prior information and observed data.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The **population parameter is fixed**, while the sample data are considered random.

15. What does long-run frequency mean in frequentist statistics?
It means that if an experiment is repeated many times, the relative frequency of an event approaches its true probability.
16. Why is random sampling important in the frequentist approach?
Random sampling helps obtain a representative sample and reduces bias, making statistical conclusions more reliable.
17. Can a population parameter be considered a random variable in the frequentist approach? Why?
No. In the frequentist approach, the population parameter is considered a **fixed but unknown value**. The sample statistic varies because different random samples can produce different results.
18. Give an example of a frequentist problem in a data science application.
A company takes a random sample of 1,000 users and estimates the average time spent on its website to infer the average usage time of all users.

C. Point Estimation
19. What is a point estimate?
A point estimate is a single numerical value calculated from a sample and used to estimate an unknown population parameter.
20. What is an estimator?
An estimator is a statistical formula or rule used to estimate a population parameter from sample data.
21. What is the difference between an estimator and an estimate?

- **Estimator:** The formula or method used for estimation.
- **Estimate:** The actual numerical value obtained from the sample.

For example, (\bar{x}) is an estimator, while **32 minutes** is an estimate.
22. What is the point estimator for the population mean?
The sample mean:

$$
\bar{x}=\frac{\sum x_i}{n}
$$

is used to estimate the population mean (μ).

23. How do you estimate a population proportion from a sample?

The sample proportion is used:

$$
\hat{p}=\frac{x}{n}
$$

where (x) is the number of successful observations and (n) is the sample size.

24. What is the point estimate of population variance?

The sample variance is used to estimate the population variance:

$$
s^2=\frac{\sum(x_i-\bar{x})^2}{n-1}
$$

25. Why is the sample mean commonly used to estimate the population mean?

The sample mean is commonly used because it is an **unbiased estimator**, easy to calculate, and becomes more reliable as the sample size increases.

26. What properties should a good estimator have?

A good estimator should ideally have:

- **Unbiasedness**
- **Consistency**
- **Efficiency**
- **Low variance**

27. What is an unbiased estimator?

An unbiased estimator is one whose expected value is equal to the true population parameter.

$$
E(\hat{\theta})=\theta
$$

28. What is meant by the bias of an estimator?
Bias is the difference between the expected value of an estimator and the true population parameter.
[
Bias(\hat{\theta})=E(\hat{\theta})-\theta
]
29. What is the difference between bias and variance of an estimator?

- **Bias** measures systematic error from the true parameter.
- **Variance** measures how much the estimator changes across different samples.

30. Why can two different samples produce different point estimates for the same population parameter?
Different random samples contain different observations, causing **sampling variability**. Therefore, their sample statistics and point estimates may differ.

D. Variability in Estimates
31. Why does a point estimate have uncertainty?
A point estimate is based on a sample rather than the entire population. Different samples may produce different estimates.
32. What is the sampling distribution of an estimator?
It is the probability distribution of an estimator or sample statistic obtained from all possible samples of the same size from a population.
33. What is the standard error?
Standard error measures the variability of a sample statistic across different samples.
For the sample mean:
[
SE=\frac{\sigma}{\sqrt{n}}
]
34. How is standard error different from standard deviation?

- **Standard deviation:** Measures variability among individual data values.
- **Standard error:** Measures variability of a sample statistic, such as the sample mean.

35. How does increasing the sample size affect the standard error of the sample mean?

As sample size increases, the standard error decreases:

[
SE=\frac{\sigma}{\sqrt{n}}
]

36. Why does variability in an estimate decrease when the sample size increases?

A larger sample contains more information about the population and reduces the effect of random sampling variation.

37. What is the relationship between standard error and confidence intervals?

The standard error determines the margin of error. A smaller standard error generally produces a narrower confidence interval.

[
CI=\text{Point Estimate}\pm(\text{Critical Value}\times SE)
]

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect?

No. Different sample means can occur naturally because of random sampling variability. Both samples may be valid.

E. Confidence Intervals

39. What is a confidence interval?

A confidence interval is a range of values calculated from sample data that is used to estimate an unknown population parameter.

40. Why is a confidence interval more informative than a point estimate?

A point estimate provides only one value, while a confidence interval provides a range and shows the uncertainty associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

It means that if the same sampling procedure were repeated many times, approximately **95% of the calculated confidence intervals would contain the true population parameter**.

42. What is the difference between a confidence level and a confidence interval?

- **Confidence level:** The percentage of confidence, such as 95%.
- **Confidence interval:** The calculated range of possible values for the population parameter.

43. What factors determine the width of a confidence interval?

The width depends mainly on:

- Sample size
- Variability or standard deviation
- Confidence level

44. How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size decreases the standard error, resulting in a **narrower confidence interval**.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

A higher confidence level requires a larger critical value, resulting in a **wider confidence interval**.

46. What is the difference between a z-confidence interval and a t-confidence interval?

- A **z-confidence interval** is generally used when the population standard deviation is known or the sample size is large.
- A **t-confidence interval** is used when the population standard deviation is unknown, especially for smaller samples.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

Use the **t-distribution** when the population standard deviation is unknown and is estimated using the sample standard deviation, particularly when the sample size is small and the population is approximately normal.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to make a decision about a population parameter based on sample data.

49. What are the null hypothesis ((H_0)) and alternative hypothesis ((H_1))?

- **Null hypothesis ((H_0)):** Represents the default assumption or no significant effect/difference.
- **Alternative hypothesis ((H_1)):** Represents a claim that there is an effect, difference, or relationship.

50. How can a confidence interval be used to make a decision about a hypothesis?

Compare the hypothesized value with the confidence interval:

- If the hypothesized value **lies outside** the confidence interval, reject (H_0) at the corresponding significance level.
- If the hypothesized value **lies inside** the confidence interval, there is insufficient evidence to reject (H_0).

Example: If a 95% confidence interval for a population mean is **(18, 22)** and the hypothesized mean is **25**, then 25 lies outside the interval, so we reject (H_0) at the 5% significance level.